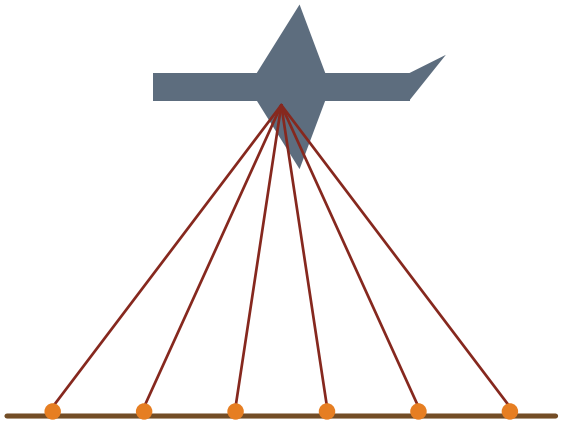
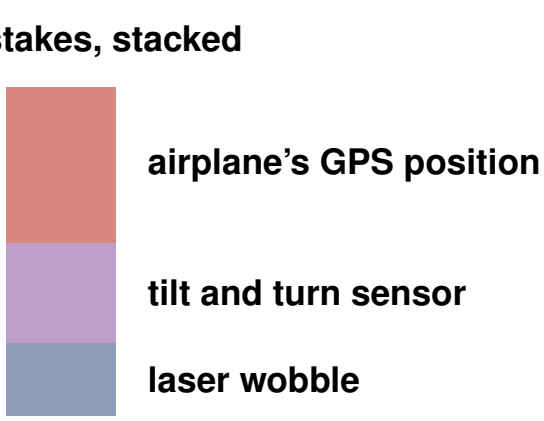
Survey-Grade Case Study: RIEGL VUX-240

BIG IDEA

With a top scanner, the biggest mistake is often where the airplane is.





THE IDEA

This scanner is a super-fast, super-careful version of the ones we met before. It fires up to 2.4 million laser flashes a second from an airplane. To know where each dot lands 500 feet below, we add up all the little mistakes: the laser's own wobble, the tiny tilt and turn errors of the motion sensor, and where GPS thinks the airplane is.

REMEMBER

- ✓ The farther away the ground, the more a tiny tilt moves each dot.
- ✓ With a scanner this good, the laser is no longer the weak link.
- ✓ Great results come from the whole system, not from the laser alone.

TRY THIS

Stack building blocks, one for each mistake: a small one for the laser, a medium one for tilt, a big one for GPS. Which block makes the stack tallest?

The full module is in the Reference Book.